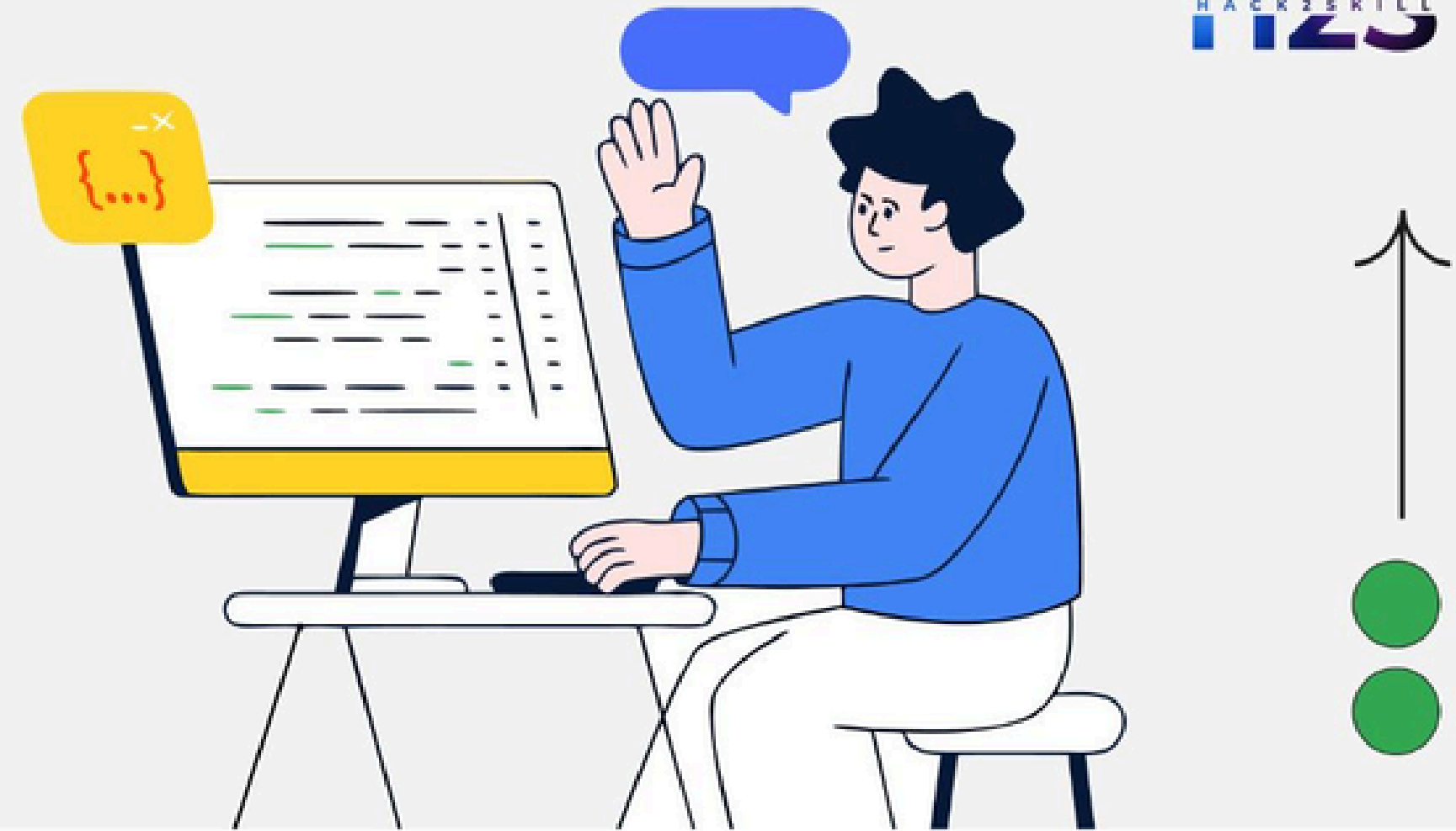


Solution Challenge



Team name: MindGlitch

Team leader name: Kashish Gupta

Problem Statement: A once-in-a-decade census and manual processes result in outdated and error-prone data. Lack of real-time tracking leaves many citizens unregistered and the system inefficient.

·∴ Brief about your solution

JanSankhya AI is India's first real-time, AI-powered Smart Census Platform. It replaces the decade-long manual census cycle with a live digital ecosystem that captures births, deaths, migrations, and demographic changes the moment they happen.

Aadhaar-Linked Registry

Every citizen linked to UIDAI Aadhaar. Camera scan auto-fills all details using Gemini Vision AI in under 3 seconds.

Dual Database Architecture

Unverified changes sit in DB2 (Change Log). Gemini AI validates each one. Only approved records enter DB1 (Population Registry).

Gemini AI at Every Step

Gemini 2.0 Flash powers the chatbot, security scanner, Aadhaar Vision OCR, and anomaly detection — all in real time.

∴ Opportunities

The following aspects define how our solution is different :

- India's census is done every 10 years manually — JanSankhya AI makes it continuous and real-time.
- No existing system combines Aadhaar scanning + Gemini Vision + dual verification databases in one platform.
- Current tools need separate apps for births, deaths, migration — we unify everything in one government-grade portal.

The solution works as follows :

- Eliminates manual errors, delays, and data inconsistencies.
- Automates data collection and verification processes.
- AI-driven validation ensures higher accuracy and fraud detection.
- Real-time cloud syncing enables instant system-wide updates.
- Combines automation, AI, and centralized data management.
- JanSankhya AI Creates a faster, reliable, and scalable population management system.

The key highlight of our solution is:

- **Real-Time Census**

Population data updates live — not once every 10 year

- **AI Security Gate**

No unverified record enters Population Registry —
Gemini checks every change.

- **Gemini Vision OCR**

Camera scans real Aadhaar card → auto-fills all fields
in 3 seconds.

- **Cloud Backbone**

Firebase Firestore stores all data permanently in
cloud — survives any device failure

∴ List of features offered by the solution



AI Census Chatbot



Aadhaar scan and auto-fill of citizen details



Gemini AI Security Check



AI-based data validation and duplicate detection



Real-time digital population data collection



Smart search (By Aadhaar, name, location)



Dual Database System



Live scalable dashboard with stats, alerts, and easy interface.

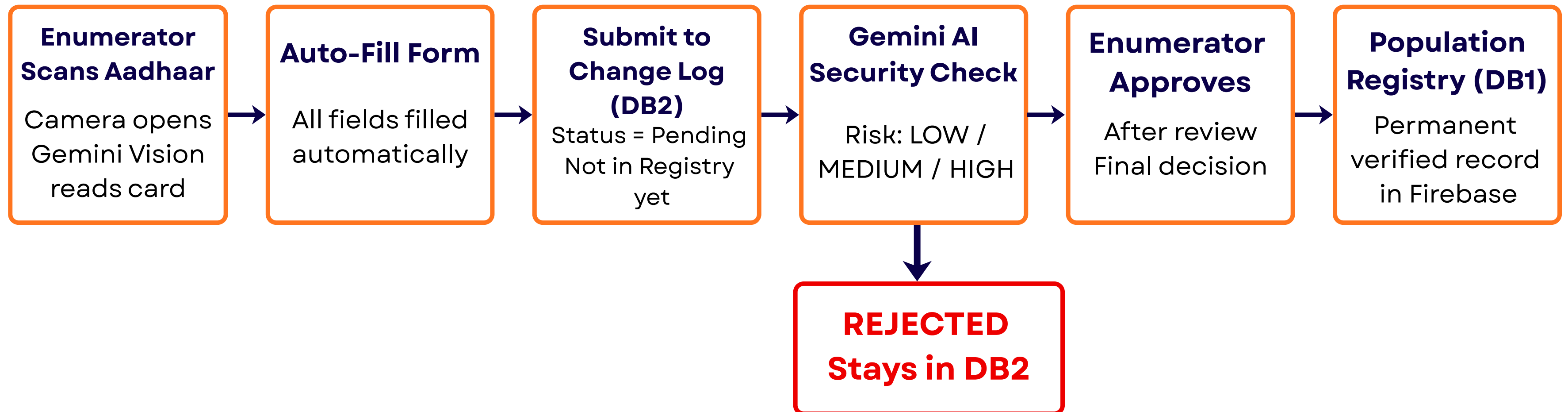


Role-Based Secure Login



Birth and death registration modules Data verification system

∴ Process flow diagram or Use-case diagram



∴ Architecture diagram of the proposed solution

Frontend Layer

- HTML + CSS + JavaScript
- Government-grade UI
- Role-based Login (3 roles)
- Live Dashboard
- Aadhaar Scanner (Camera-based)
- Dual Database Tables
- AI Chatbot Interface
- Field Report Forms

AI Layer (Google Gemini)

- Gemini 2.0 Flash(Chatbot)
- Gemini 1.5 Flash(Image Processing)
- Aadhaar OCR Scanner
- Security Risk Analysis
- Census AI Chatbot
- Change Verification System

Database Layer (Firestore)

- Firestore DB1(Population Registry)
- Firestore DB2(Change Logs)
- Firestore Collection (Newborn Records)
- Real-time Listeners
- Cloud Deployment

Data Flow : Frontend → Gemini API → Firestore → Real-time Update → Dashboard

∴ Technologies to be used in the solution

Frontend:

- HTML5 + CSS3
- Vanilla JavaScript
- Canvas API (for graphs & charts)
- Google Fonts (Noto Sans / Syne)
- Government-style UI design

Cloud Backend:

- Real-time data sync (onSnapshot)
- Dual data collection architecture
- Cloud deployment
- Firebase Firestore
- Firebase authentication

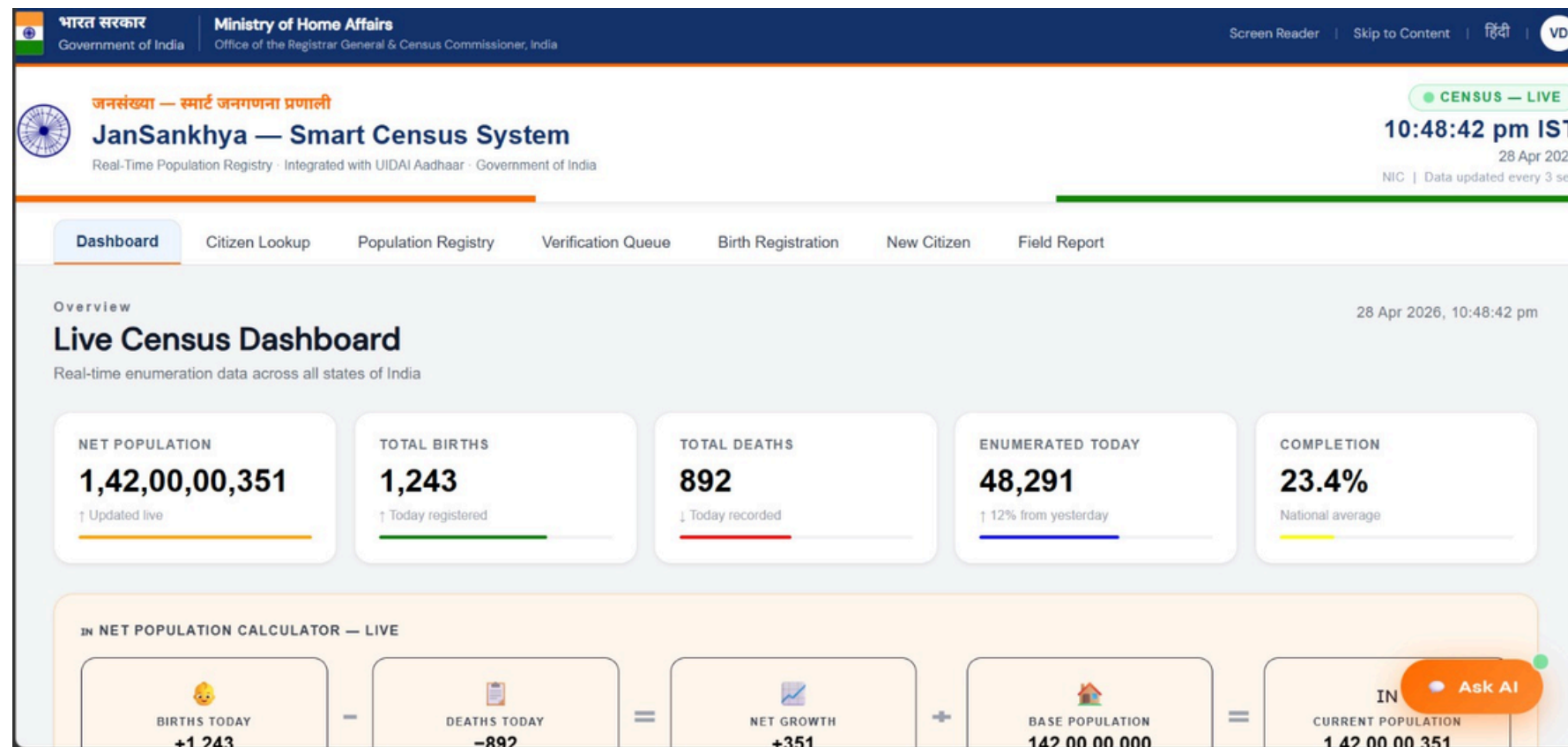
AI & Intelligence:

- Gemini 2.0 Flash(Chatbot)
- Gemini 1.5 Flash (Image processing)
- Aadhaar Card OCR (text extraction)
- Security risk analysis (AI-based)
- Real-time AI responses

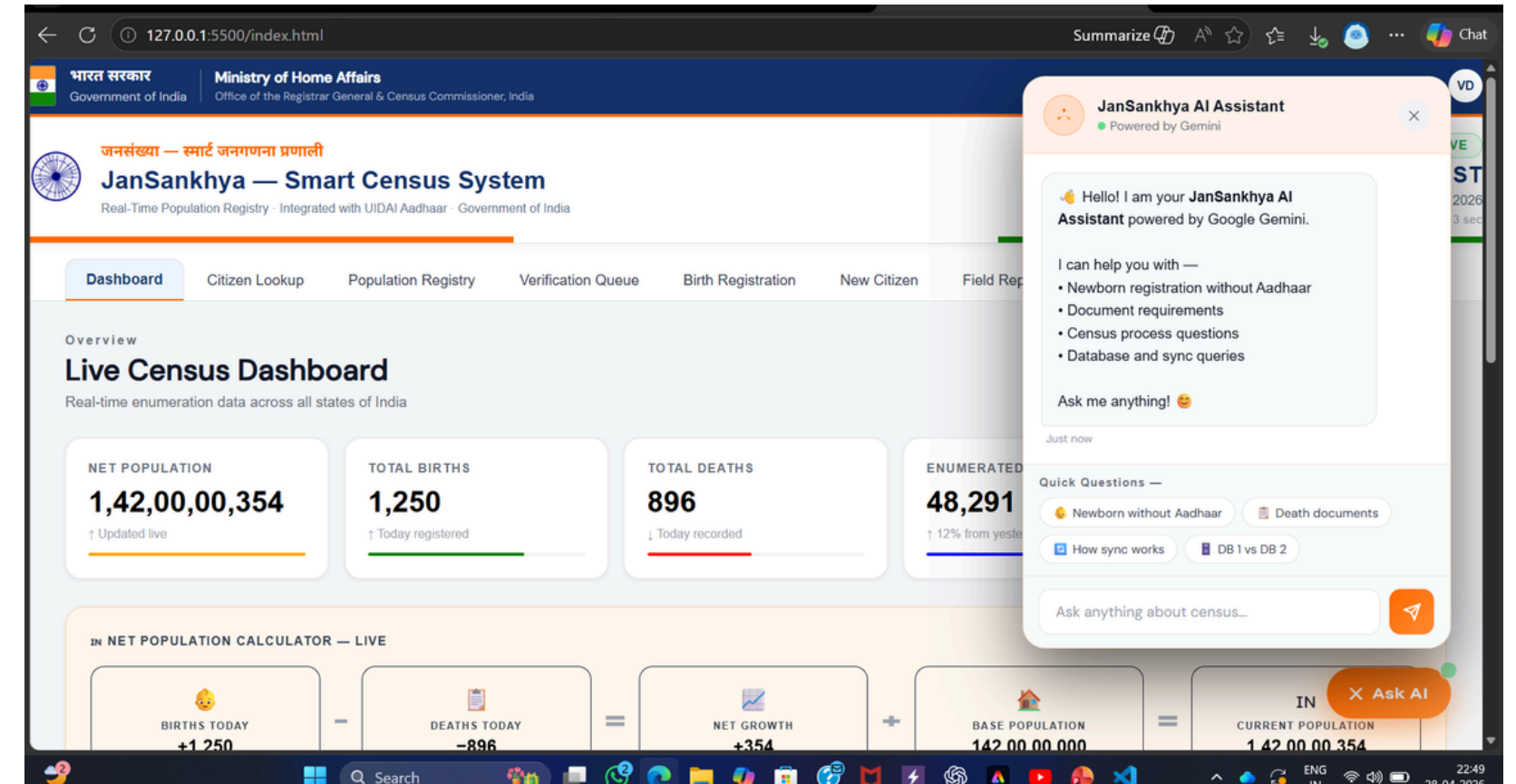
APIs & Integration:

- Gemini REST API
- Firebase JS SDK v9
- Browser Media
- Devices API (camera access)
- UIDAI Aadhaar Framework
- Web Camera Access API

Snapshot of the MVP



Live Dashboard



AI ChatBot



Solution Challenge



भारत सरकार Ministry of Home Affairs
Government of India Office of the Registrar General & Census Commissioner, India

जनसंख्या — स्मार्ट जनगणना प्रणाली
JanSankhya — Smart Census System
Real-Time Population Registry · Integrated with UIDAI Aadhaar · Government of India

LIVE SYNC ACTIVE
10:53:38 pm IST
28 Apr 2026
NIC | Data updated every 3 sec

Home About Census 2027 Self-Enumeration Resources State Offices Contact Us RTI

Home > Official Login — Enumerator / Supervisor / Admin

Census 2027

India's First Fully Digital Census

- Phase 1 — House Listing Operations
April 2026 to September 2026
- Phase 2 — Population Enumeration
February 2027 · Reference date March 1, 2027
- Mobile App Based Data Collection
30+ lakh enumerators across India
- Aadhaar Integrated Auto-Fill
Linked with UIDAI for instant citizen lookup
- Secured by Gemini AI
Every change verified before database update
- Pan-India Coverage
6.4 lakh villages · 5,128 statutory towns

This is a prototype demonstration system for the 2027 Digital JanSankhya initiative. For official data, visit censusindia.gov.in

Enumerator / Official Secure Login

जनगणना कार्यकर्ता / अधिकारी सुरक्षित लॉगिन

SELECT YOUR ROLE

Enumerator Supervisor Admin

Enumerator: Can scan Aadhaar, collect household data, register newborns, and submit changes in their assigned zone.

Employee ID / Enumerator Code
e.g. ENUM-GJ-2847

State / Union Territory
— Select State —

Password
Enter your password

Security Captcha
Q Z F L Z H Type captcha here

Login Securely to JanSankhya 2027

Forgot Password? Register / Activate Account Technical Help

DEMO LOGIN CREDENTIALS (FOR TESTING ONLY)

Enumerator ID: ENUM-GJ-2847 Pass: enum123 Click to auto-fill	Supervisor ID: SUP-GJ-001 Pass: super123 Click to auto-fill	Admin ID: ADMIN-001 Pass: admin123 Click to auto-fill
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127.0.0.1:5500/index.html

ENUMERATOR TOOL

Aadhaar Smart Scanner

Point camera at Aadhaar card — Gemini Vision AI reads and fills all details automatically

Show real Aadhaar card to camera — Gemini AI reads name, number, DOB, address and saves directly to Population Registry on confirmation.

Tap to Open Camera
Hold real Aadhaar card steady in frame

OR UPLOAD IMAGE

Upload Aadhaar image

SCANNING TIPS

- Good lighting — no shadows on card
- Hold card flat, all 4 corners visible
- Keep card as close to camera as possible
- Avoid blurry or dark captures

Scan an Aadhaar Card
Extracted details appear here after scanning

Ask AI

Aadhaar Scan Screen

Login Page

∴ Additional Details/Future Development (if any)



Official UIDAI Aadhaar API integration



Self-enumeration portal for citizens



Multi-language support



Integration with civil registration systems



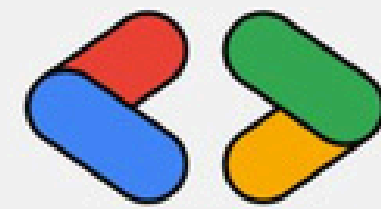
Offline-first PWA for rural enumerators



National Population Register (NPR) sync

∴ Provide links to your

1. GitHub Public Repository : <https://github.com/vedanshi2209/Jansankhya-ai>
2. Demo Video Link (3 Minutes) :
https://drive.google.com/file/d/1YT9x8zkkin79H8E4Ber0o7LhC4Icw_Cm/view?usp=drive_link
3. MVP Link : mindglitch123.netlify.app
4. Working Prototype Link : mindglitchteam.netlify.app



Solution Challenge



Thank You